Daniel Tymecki

AI PRODUCT ENGINEER · FORWARD DEPLOYED ENGINEER · FULL-STACK

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SUMMARY

Forward-deployed AI product engineer who turns ambiguity into shipped product. Embeds with portfolio managers, distills complex research workflows into agentic AI tools, and ships them to production in days, not quarters. Pairs clear communication and product taste with end-to-end ownership, from the first conversation to design, full-stack build and deployment at scale.

SELECTED IMPACT

> **541 merged pull requests across 25 repositories** since 2025, spanning web apps, Python services, AI agent skills and deployment infrastructure.

> **0 → 1 → 100+:** designed and shipped an agentic AI workspace to production; users grew **42% in three weeks**, ahead of a business-wide launch.

> **9 weeks** to deliver the research portal for a newly launched investment team. The team's head rated it above anything at his prior firm and later called the platform the best in the industry; department leadership credited the build as a *"one-man show."*

> **350+ code reviews for 27 engineers in 10 months** (~60% of the team's frontend changes), acting team lead during manager absences, mentor to early-career engineers and interns, and interviewer across **6 hiring loops**.

EXPERIENCE

Millennium · Global multi-strategy hedge fund · Miami, FL — Jan 2023 – Present

AI Engineer — Equities Technology, Applied AI — 2025 – Present

- **Rebuilt and led the Equities AI web platform:** re-architected the group's AI research web platform on Next.js/React and became its primary engineer and code owner (~950 commits). It provides single sign-on with role-based access and impersonation, streaming LLM chat, AI workspaces, alerting, and earnings and research dashboards, all built on a versioned shared component library.
- **Agentic AI workspace (0 → 1):** designed and built the web application for an agentic research workspace on a Python agent backend. Features include live multi-agent sessions, scheduled monitoring workflows with a calendar view, a skills and plugin marketplace, generated dashboards, and email/OneNote integration via Microsoft Graph.
- **Forward-deployed delivery:** embedded with a newly launched investment team to build a research platform for expert-network transcripts in about nine weeks, shipping to production between weekly sessions with the portfolio manager. It covers semantic, keyword and structured search; collaborative annotations on a production annotations backend; find-in-document over PDFs; and quote-finding AI chat. Also contributed to executive demos for the firm's global co-heads of equities.
- **Generative UI and agentic chat:** built a pipeline in which LLMs convert long-form research into structured, interactive widgets that stream to the browser (AG-UI protocol), with LLM reflection/validation and source citations. Rebuilt the platform's research chat around specialist agents (prediction markets, company experts, M&A), with concurrent streaming, inline citations and Glean enterprise-search integration.
- **AI research tooling:** shipped a generative research grid in which an LLM answers each ticker-and-question cell with citations, with shareable sessions. Also built document structured-extraction and annotation tools, sentiment, narrative and keyword-trend widgets, and a unified alert center.
- **Full-stack services:** built Python/FastAPI services for streaming report transformation, market prices (Redis-cached, with global ticker lookup), role-gated annotations, earnings calendars and usage analytics on SQL Server. Owned the Next.js server tier: SSO, Microsoft Graph OAuth and stateless signed-token sessions.
- **Deployment at scale:** migrated services from HashiCorp Nomad to Kubernetes and deployed across dev, UAT and prod on on-prem Kubernetes and GKE (Helm, Argo CD, GitHub Actions, health checks, runtime configuration). Created an app template that launches a secure, deployable UI in five commands; a peer used it to launch a new product.
- **Security and compliance:** enforced market-data licensing (e.g., Bloomberg) across AI tools, using role sync from verified workstation connectivity and Kerberos end-user impersonation. Gated licensed news and alternative data by entitlement.
- **Technical leadership and design:** named Tech Lead in the department's AI product-ownership plan, and code owner of four codebases. Authored 24 architecture proposals, the team's design and practice guide, and the merge policy for AI code-review agents. Led UX from prototypes built in code through to production, and created a brand design system, with automated conformance tests, adopted by nine AI agent skills.

- **Mentorship and hiring:** alumnus of Millennium's early-career engineering program (LEaD) who now mentors its rotation engineers and the team's summer intern: scoping first projects, onboarding them to the codebase and reviewing their code. Oversaw interviews for several different roles.

Full Stack Software Engineer — Finance Technology (Tech Lead) Jan 2023 – 2025

- **Tech-led a cloud-based portfolio-management tooling suite:** coordinated front- and back-end engineers and partnered with quantitative risk, senior management and portfolio management users to streamline operational workflows.
- **Built shared Python and Angular platform libraries used across the suite:** auto-generated OpenAPI/Swagger docs, Postgres reference-data access with filtering and selectors, Excel exports, email and report templating, and SSO token auto-refresh. Upgraded the web layer to Angular 16 with Jenkins CI.

L3Harris Technologies · Aerospace and defense technology · Palm Bay, FL Jan 2020 – Jan 2023

Senior Associate Software Engineer II Nov 2021 – Jan 2023

- Promoted to lead an integrated product team building a cloud platform for autonomous antenna requisitioning across client missions.
- Architected a configurable, multi-threaded Python simulation-engine microservice that helped win a **$500M+** business pursuit.
- Advised senior systems engineers on microservice architecture and software practices for a **$200M** software redesign.
- Led engineer onboarding and training (Git, VS Code, Docker) and served as lead panel interviewer for hiring.

Associate Software Engineer I Jan 2020 – Nov 2021

- Built microservices for a cloud-based distributed system that plans, schedules and executes space missions.
- Developed faster-than-real-time C++ flight software for satellites, in collaboration with astrophysicists.
- Built a server-side JavaScript caching layer that cut load times **24%**, and designed Astro UX-compliant React components.
- Introduced project-wide testing standards, raising code coverage **33%**.

RECOGNITION

- **Featured engineer, Millennium's *Creative Engineer* series** (2026) — Public Q&A profile on building agentic AI tools for investment teams, published at mlp.com.
- **Credited as a *"one-man show"* by department leadership** (2026) — Recognized after a new investment team's head rated the research portal above anything at his previous firm.
- **Named Tech Lead in the Equities AI product-ownership plan** (2026) — Also recognized in leadership's launch note for the agentic workspace rollout.

SKILLS

- **AI & LLM:** LLM application development; agentic systems (tool use, planning, memory, reflection); retrieval-augmented generation (RAG); semantic search and reranking; prompt engineering; model selection; LLM output validation; generative UI (AG-UI); Model Context Protocol (MCP); Claude Code skills; OpenAI, Anthropic and xAI models
- **Languages:** Python, TypeScript, JavaScript, SQL, C++, Java, C#, HTML/CSS, Bash
- **Frontend:** React 19, Next.js 16, Angular, Tailwind CSS, shadcn/ui, Zustand, TanStack Query, Zod, PDF.js, Recharts, Cytoscape, Playwright, Vitest
- **Backend & Data:** FastAPI, Flask/Falcon, Node.js, REST/OpenAPI, server-sent events, async and event-driven systems, microservices, Kafka, RabbitMQ, PostgreSQL, SQL Server, MongoDB, Redis, SQLAlchemy, Pydantic, pytest
- **Cloud & DevOps:** Kubernetes (GKE, on-prem), Helm, Argo CD (GitOps), Docker/Podman, GitHub Actions, Jenkins, GCP, AWS, HashiCorp Nomad, Datadog, Linux
- **Security & Identity:** SSO, OAuth 2.0 (Microsoft Entra, Ping), JWT, role-based access control, Kerberos delegation, data-entitlement controls
- **Product & Design:** UX and interaction design, rapid prototyping, design systems and tokens, accessibility (contrast), data visualization, stakeholder discovery, Agile delivery

EDUCATION

University of Florida — Herbert Wertheim College of Engineering · Gainesville, FL Dec 2019

B.S.E., Computer Science Engineering · Minor in Entrepreneurship

Computer Science Teaching Assistant (2018–2019): co-developed curriculum and C++, Java and Python assignments.